

Algebra II with Trigonometry

Chapter 6.1

Olympiad Problems

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Problems

1. Let t be a real number such that the terminal point determined by t on the unit circle has rational coordinates (x, y) . If t is not a multiple of $\frac{\pi}{2}$, show why both x and y must be of the form $\frac{a}{c}$ and $\frac{b}{c}$ for integers a, b, c with $a^2 + b^2 = c^2$. Give an example of such a point and determine the reference number for this t .
2. The terminal point of t on the unit circle is (x, y) . Quickly determine, without a calculator, the terminal point for $5\pi - t$ and explain the symmetry involved.
3. Consider the sequence $t_n = \pi + (-1)^n \frac{\pi}{6}$ for $n = 1, 2, 3, \dots$. For which values of n does the terminal point determined by t_n lie on the coordinate axes? Justify your answer mentally.
4. Estimate whether the sum $S = \cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right)$ is positive, negative, or zero, and explain your reasoning using the symmetry of the unit circle.
5. Suppose you start at $(1, 0)$ on the unit circle and travel a total distance of 8π in the clockwise direction. What point do you land on? Give your answer in coordinates and explain the pattern used.
6. Let t be a real number such that the terminal point on the unit circle is (x, y) . If t and $-t$ have the same reference number but different terminal

points, describe fully the relationship between their coordinates, and give a nontrivial example.